

Fundamentals of Armor Operations

INSTRUCTOR PROFILE



PERSONAL:

- Name: MAJ ROD RYAN L VIBAR (ARM) PA
- Home: PIODURAN ALBAY
- Civil Status: HAPPILY MARRIED
- Source of Commission: OPC CL 61-13

ASSIGNMENTS:

- 1Cav Bn – EXO/PL/S1
- 3rd Mech bn – EXO/S1/S4
- 2nd Mech Bn – Coy Cmdr 22nd Mech/Operation Officer
- TAS – Head Training Support Department - Current
- TAS – Chief Operation and Security Section - Current

MILITARY SCHOOLINGS:

- Adjutant General Service Officer Basic Course
- Armor Officer Advance Course
- Basic Airborne Course
- Instructor Development Training

Need

Students need to understand the Fundamentals of Armor Operation in order to effectively integrate, command, and support armored units in modern combined arms operations, enhancing battlefield mobility, firepower, and force survivability.

DURATION

3 hours

Roadmap

EXPLAIN:

- ❑Fundamentals of Armor Operation
- ❑Characteristics and Organization of Armor Units
- ❑Planning Considerations
- ❑Capabilities

APPLY: Ask a question for students (Check on Learning)

SUMMARIZE: Let the students summarized the session by defining the important data on Armor Operations.

ASSESS: The students should be able to take the written exam.
Type of exam is multiple choice with a minimum of 20 items equivalent to 100%.

Learning Objectives

At the end of the lesson the student should be able to discuss the fundamentals of Armor Operations in accordance with the Armor Operation Manual.

1. Illustrate the organization of Armor Units
2. Determine the capabilities and Limitations of Armor Assets
3. Determine the planning considerations in the employment of Armor Assets.

Classroom Rules



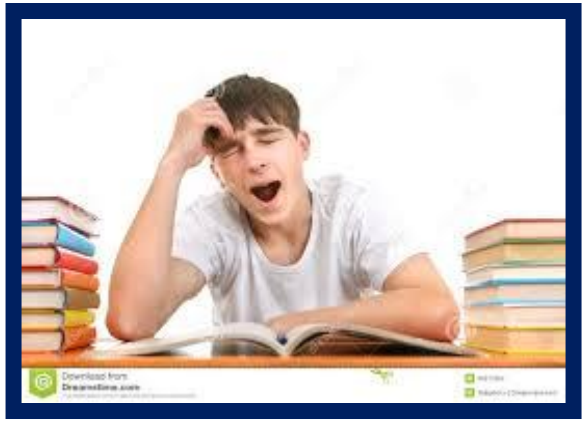
Exercise academic freedom.

Please go out silently to attend personal necessity.

Stay yourself awake

Mobile phones should be in silent mode.

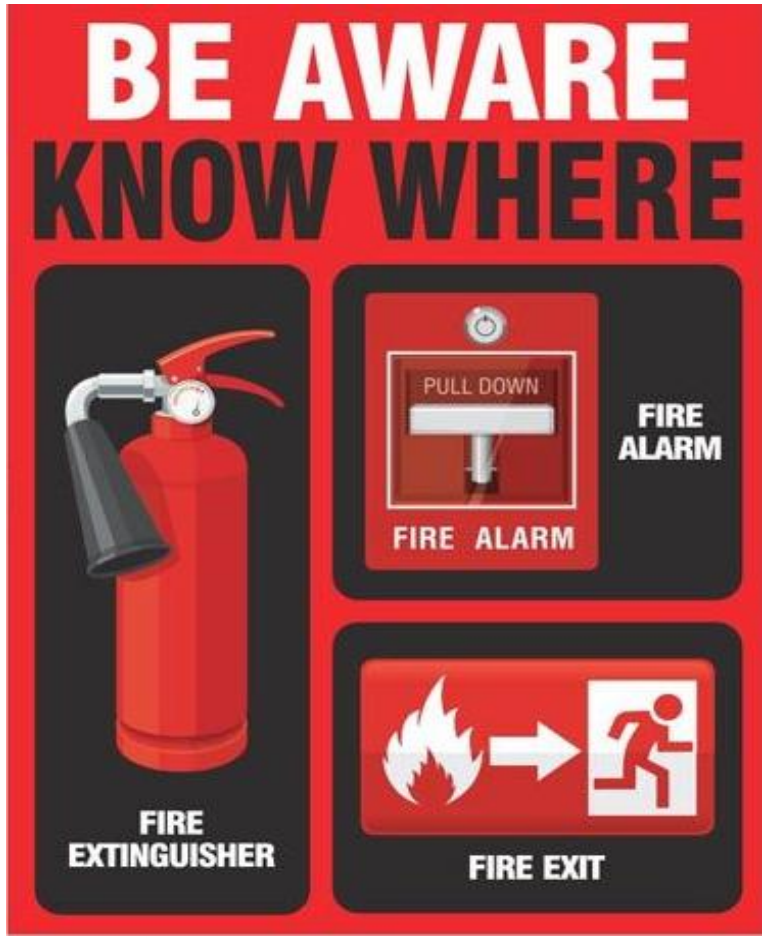








Safety Precaution





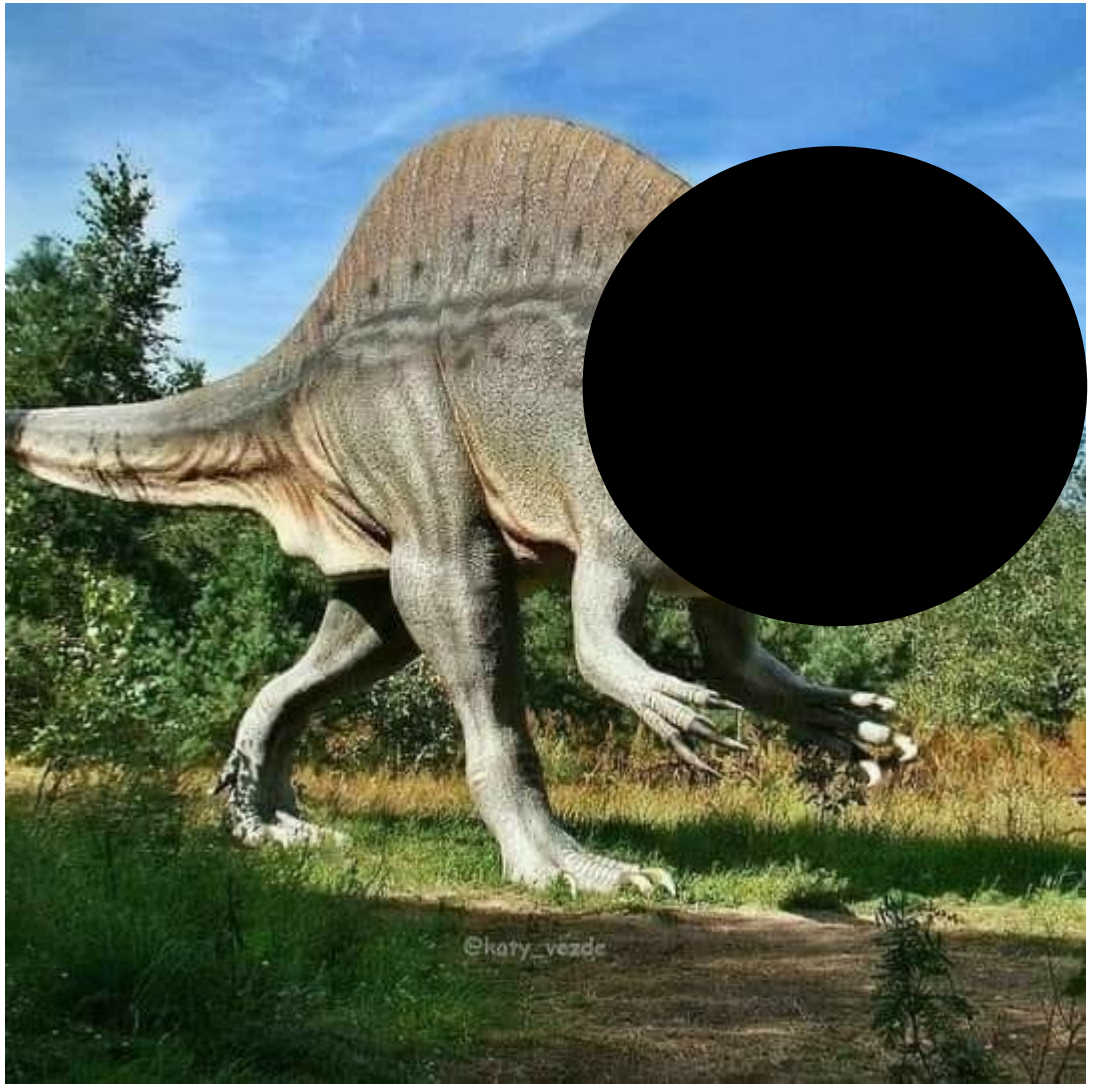


References

- ARMOR OPERATIONS MANUAL (PAM 3-0502)
- TANK OPERATIONS MANUAL (PAM 3-0502-01)
- CAVALRY OPERATIONS MANUAL (PAM 3-0502-02)
- MECHANIZED INFANTRY OPERATIONS MANUAL (PAM 3-0502-03)

Icebreaker







Icebreaker

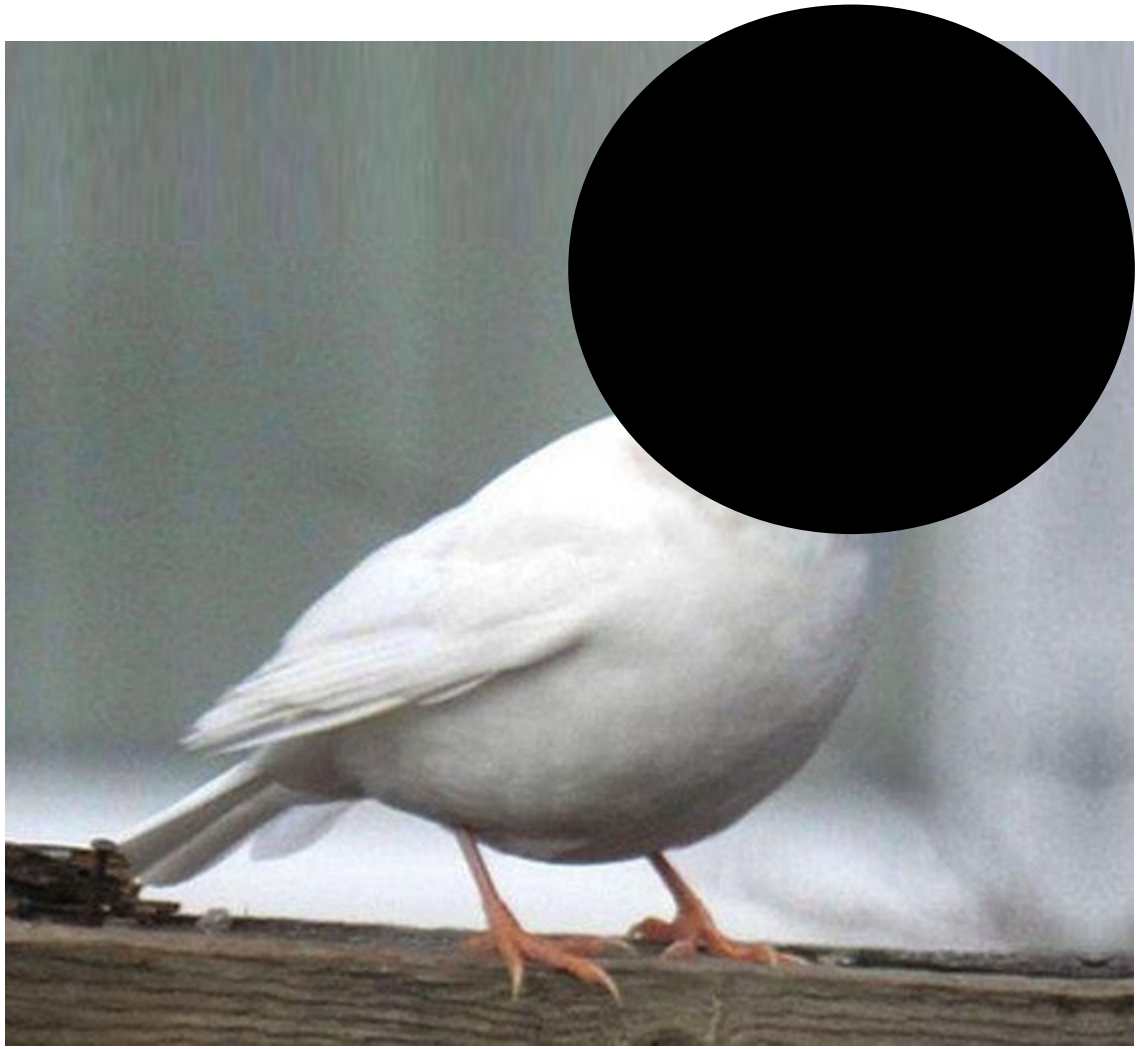






Icebreaker







FUNDAMENTALS OF ARMOR OPERATIONS

ARMOR

- is a **combined arms** force designed to conduct mounted combat employing armor-protected vehicles as a principal means of accomplishing a ground combat mission

- A defensive covering that affords protection

ARMOR OPERATION

- conducted by fire and maneuver and are combined and controlled to create a preponderance of combat power that culminates in a powerful and violent action at the decisive time and place

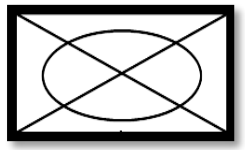
ARMOR MISSION (DOCTRINAL)

To close with and neutralize enemy forces by fire, maneuver and shock effect in coordination with other arms. They may also conduct Internal Security Operations (ISO) activities, independently or as part of a task force in peacetime environment as required by higher headquarters.

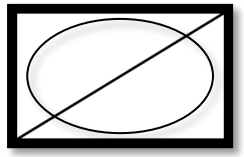
-PHILIPPINE ARMY MANUAL 3-0502 (ARMOR OPERATIONS)

FUNDAMENTALS OF ARMOR OPERATIONS

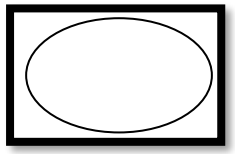
Three (3) sub-components/capabilities of Armor in the Philippine Army:



1. Mechanized Infantry Operations

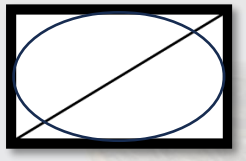


2. Cavalry Operations



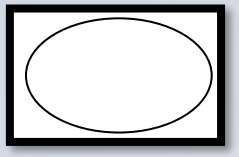
3. Tank Operations

FUNDAMENTALS OF ARMOR OPERATIONS



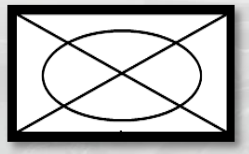
Cavalry - performs reconnaissance, surveillance and provides security for the unit to which is assigned or attached.

FUNDAMENTALS OF ARMOR OPERATIONS



Tank - neutralize enemy forces by fire, maneuver and shock effect in coordination with other units.

FUNDAMENTALS OF ARMOR OPERATIONS

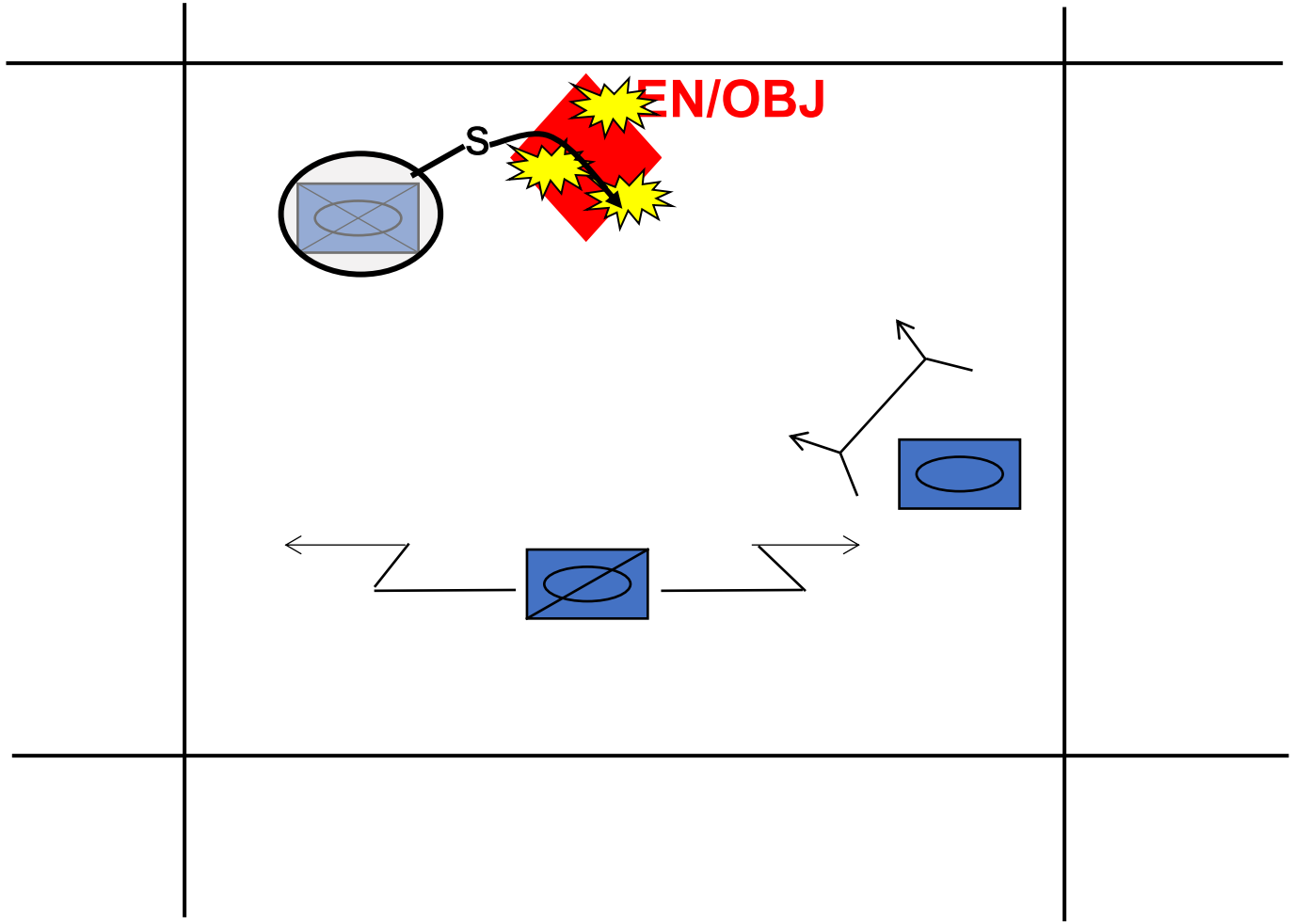


Mechanized Infantry - can close with the enemy by means of fire and maneuver to destroy or capture the enemy or to repel enemy assault by fire, close combat, and counter attacks.

ARMOR OPERATIONS



ARMOR OPERATIONS



CHARACTERISTICS OF ARMOR

- ❑ Armor Protection
- ❑ Firepower
- ❑ Mobility
- ❑ Flexibility
- ❑ Intensive and Flexible Communication
- ❑ Shock Action

MECHANIZED INFANTRY



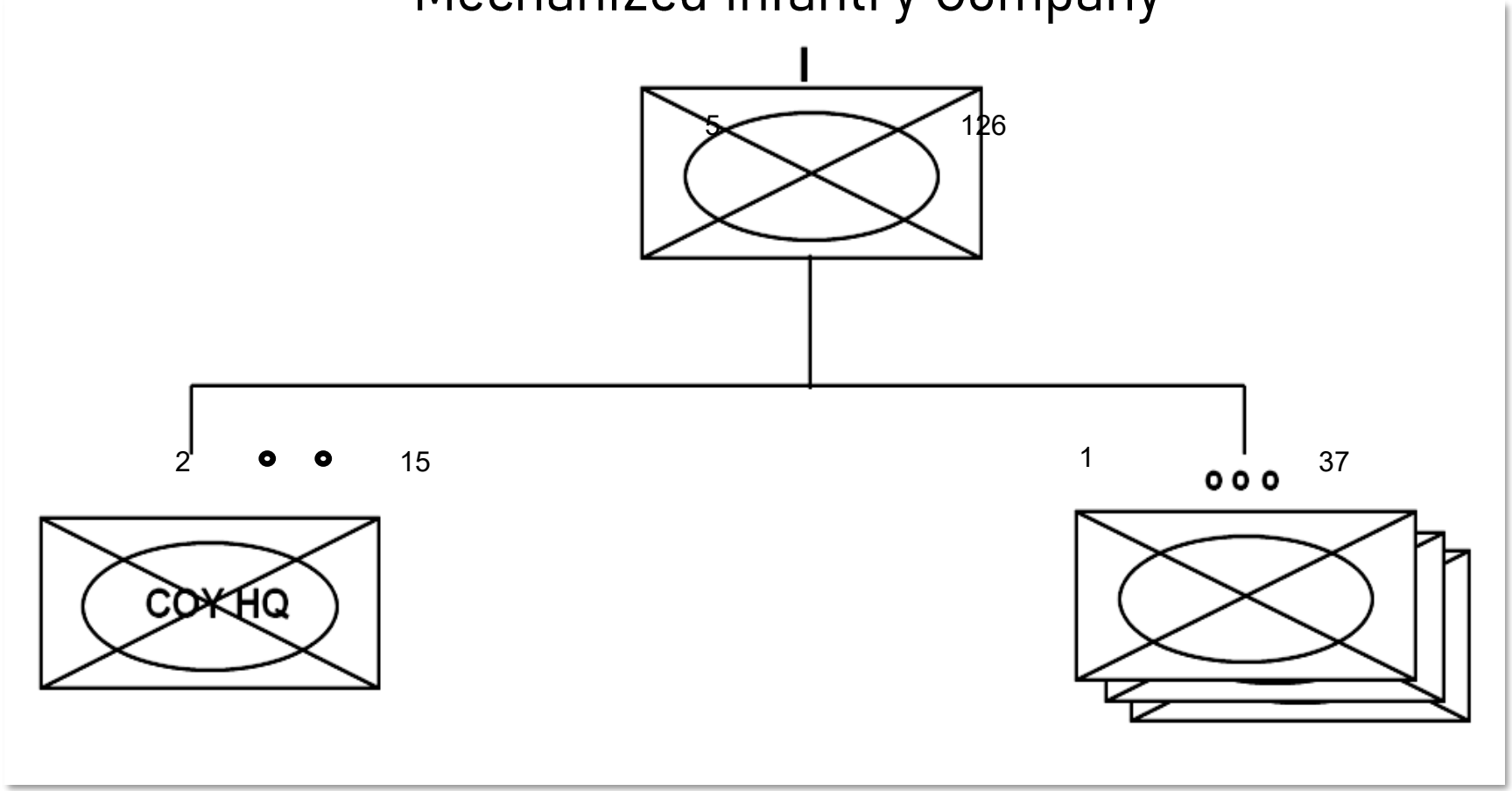
MECHANIZED INFANTRY

MISSION

The mission of the Mechanized Infantry unit is to close with the enemy by means of fire and maneuver to destroy or capture him or to repel his assault by fire, close combat, and counter-attack.

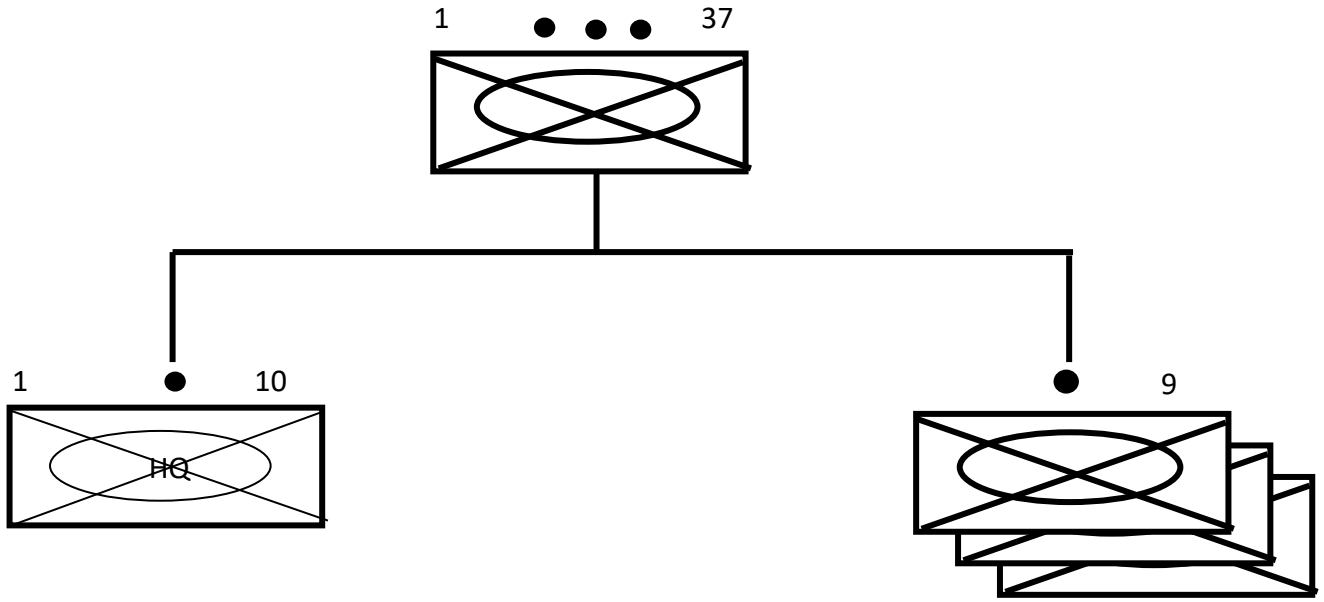
ORGANIZATION

Mechanized Infantry Company



ORGANIZATION

Mechanized Infantry Platoon



SQUAD ORGANIZATION

- The carrier team is normally composed of the carrier team leader, gunner, and driver. It functions as the vehicle crew responsible for the AIFV and its armaments. The dismount team is made up of all squad members not part of the carrier team.

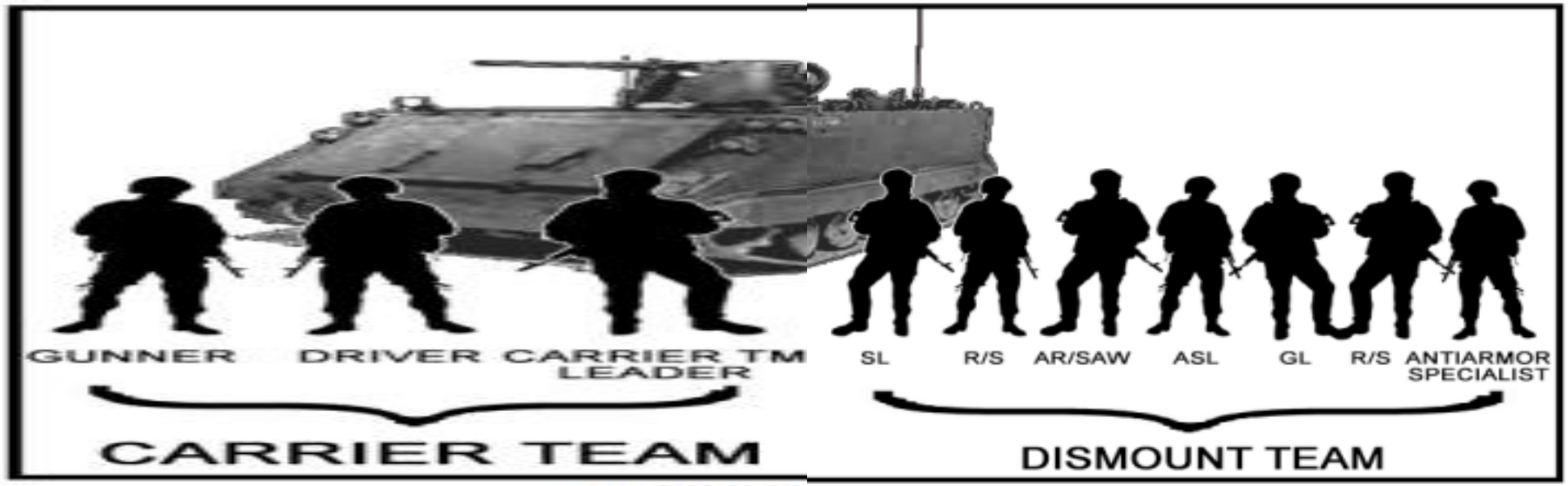


Figure 1 - Mechanized Infantry Squad

PLATOON ORGANIZATION

- The mechanized infantry platoon is equipped with four AIFVs. It is organized with a platoon headquarters (Pltn Hqs) and three mechanized infantry squads. The platoon leader and the headquarters are mounted in one AIFV while the other three squads are mounted on the other three AIFVs.



CAPABILITIES



- **WHEN MOUNTED**
  - ❑ destroy light armor vehicles
  - ❑ Suppress & kill dismounted elements
  - ❑ Secure terrain
  - ❑ Maneuver with tanks
  - ❑ Assault unprepared, suppressed defenses

CAPABILITIES



- **WHEN DISMOUNTED**
  - ❑ fix the enemy to allow maneuver
  - ❑ Infiltrate enemy positions
  - ❑ secure obstacles to defeat enemy breaching efforts
  - ❑ Move over terrain not trafficable by vehicles
  - ❑ defend vs dismounted attack

LIMITATIONS



- AIFVs and APCs are more vulnerable than tanks
- Dismounted infantrymen are vulnerable to small arms & indirect fire
- Infantrymen needs sufficient time to mount and dismount
- AVs are restricted to certain Avenues of Approach due to limitation in the terrain

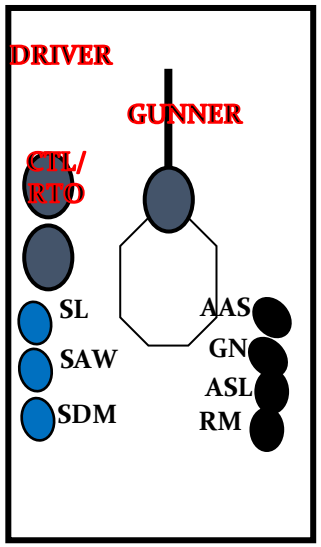
SOPs ON DISMOUNTING

- Chamber empty, muzzle down
- First to dismount is the last to mount and vice versa
- Avoid piling up at the door of the AV
- Speed is of the essence
- SL or PL is responsible for both the maneuver of the dismount team and in positioning the AV
- No riders on top of the AV
- Never hide behind an AV



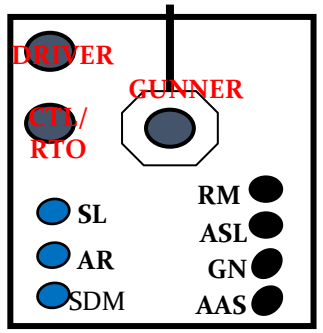
DISMOUNT

- No enemy contact yet
- Terrain is favorable
- Fast dismount is necessary

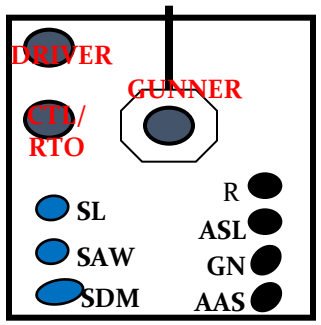


DISMOUNT LEFT

- No enemy contact
- Terrain is not favorable on the right

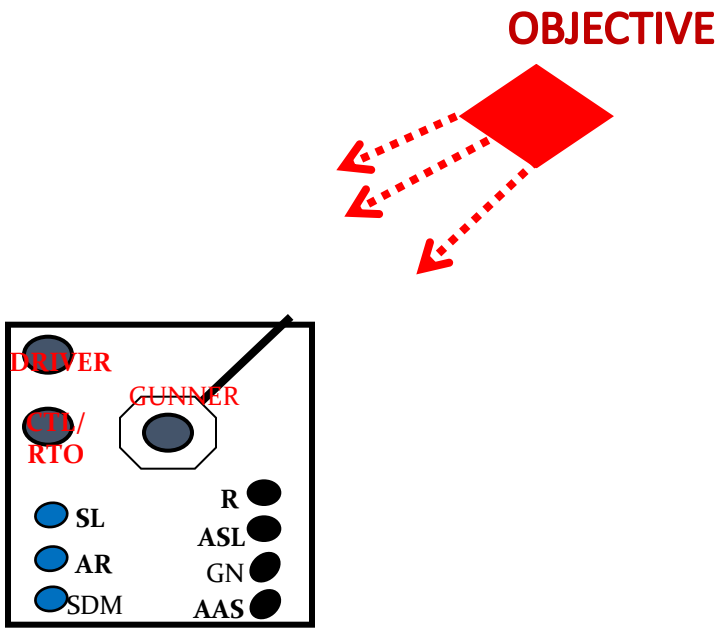


DISMOUNT RIGHT



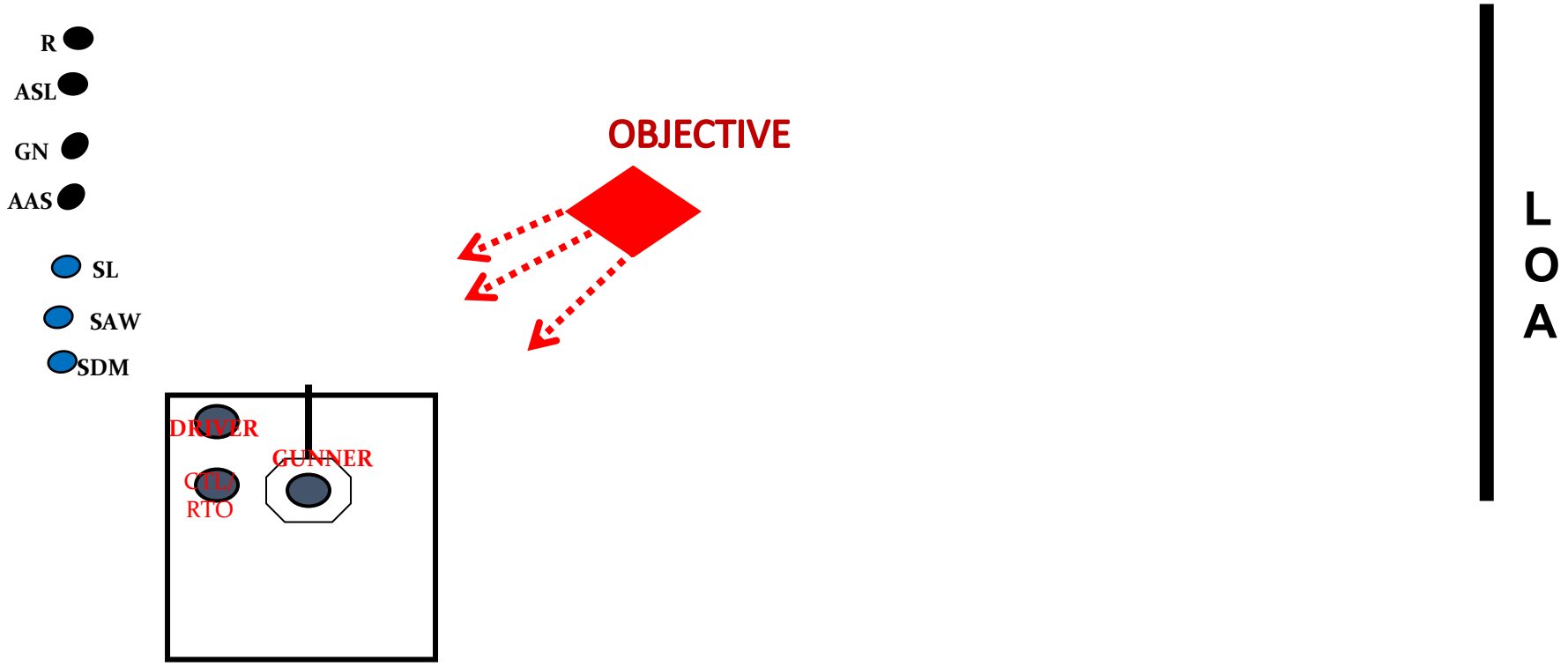
- No enemy contact
- Terrain is not favorable on the left

DISMOUNT LEFT
(under fire from right)

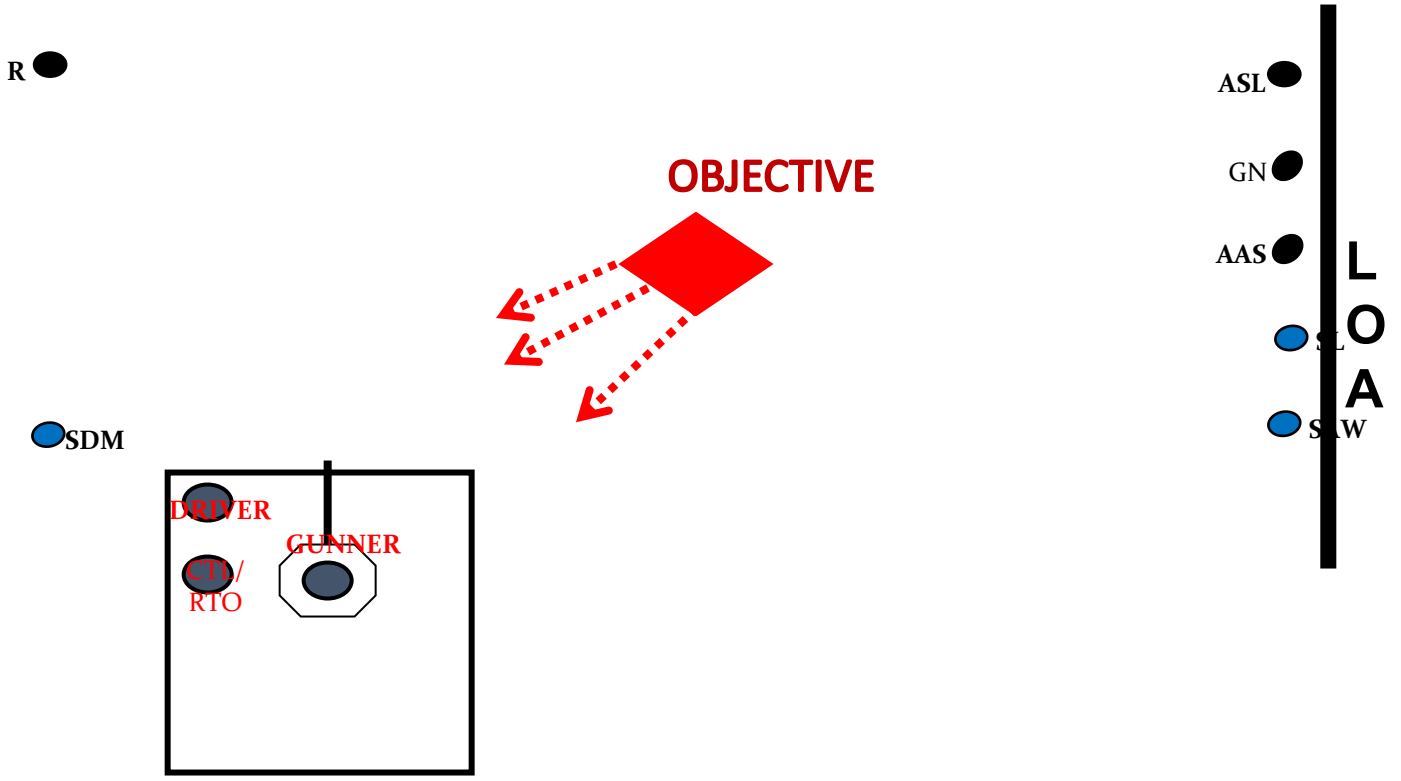


- React to contact on the right
- Extricate out of the kill zone
- Perform IAD after dismount procedures

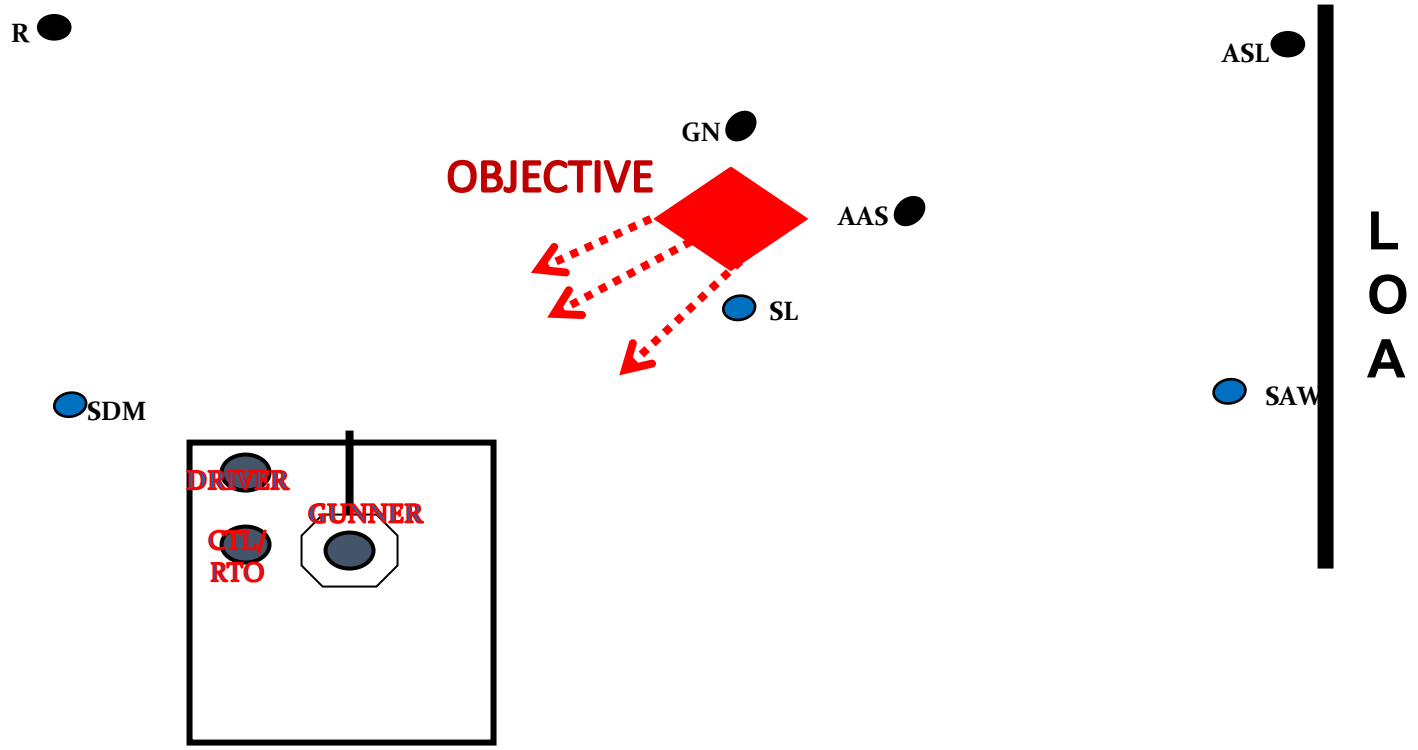
DISMOUNT LEFT
(under fire from right)



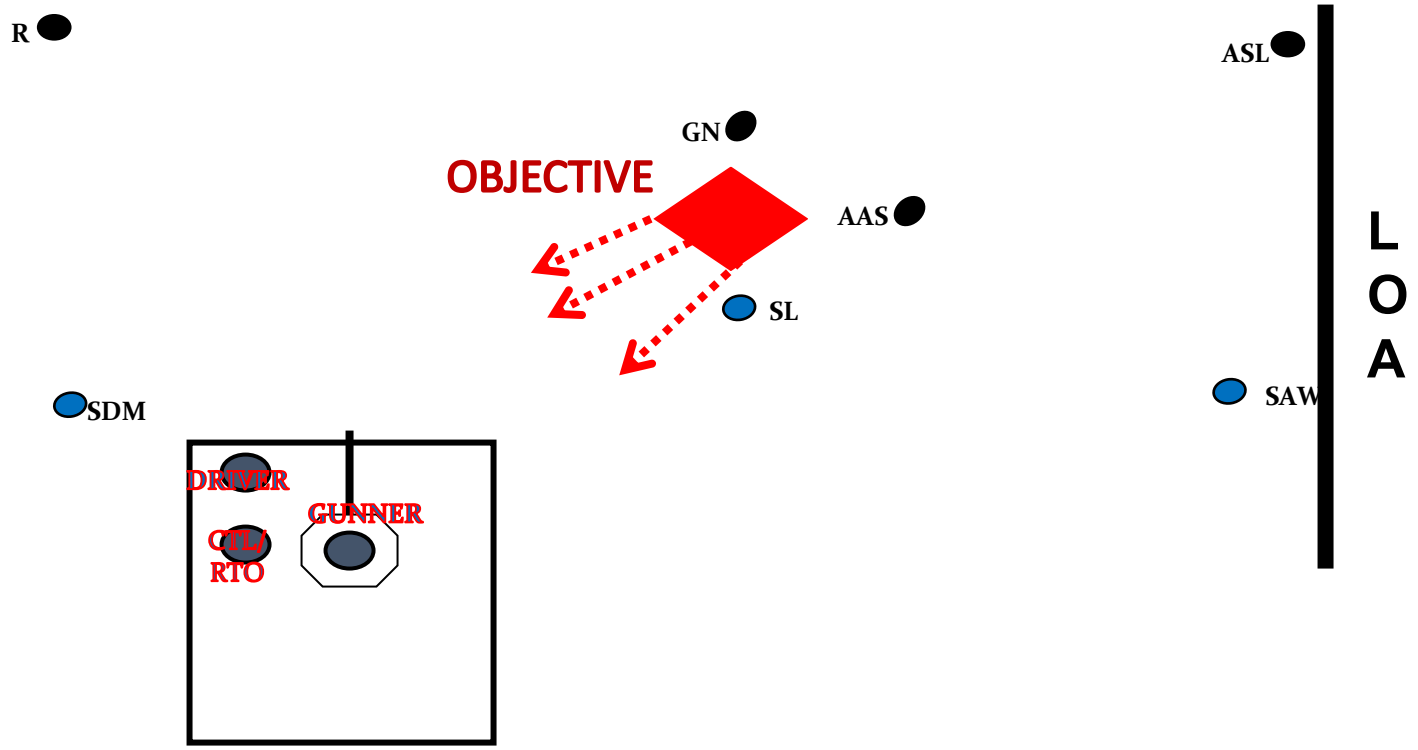
DISMOUNT LEFT
(under fire from right)



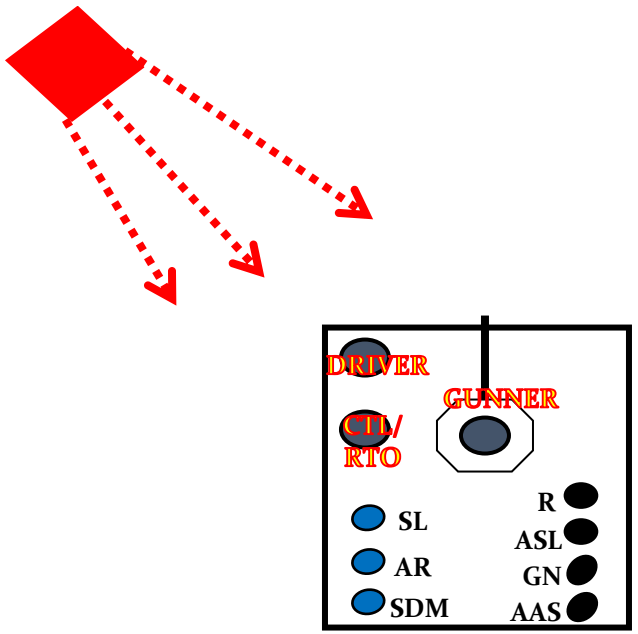
DISMOUNT LEFT
(under fire from right)



DISMOUNT LEFT
(under fire from right)

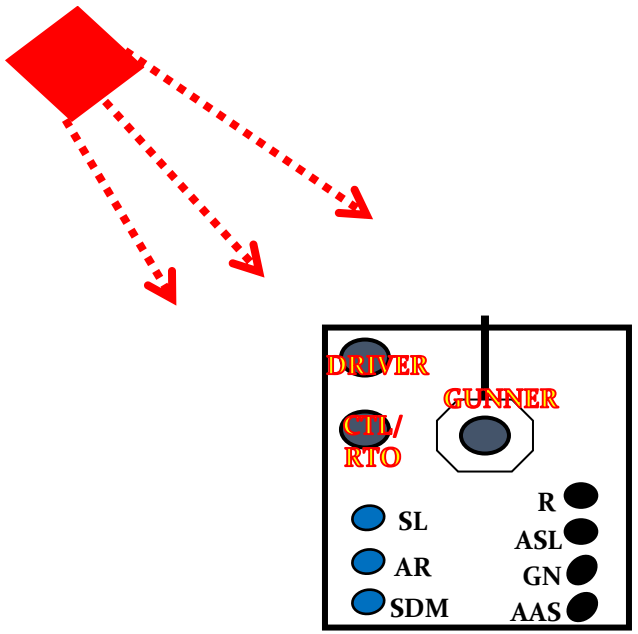


DISMOUNT RIGHT (under fire from left)



- React to contact on the left
- Extricate out of the kill zone
- Perform IAD after dismount procedures

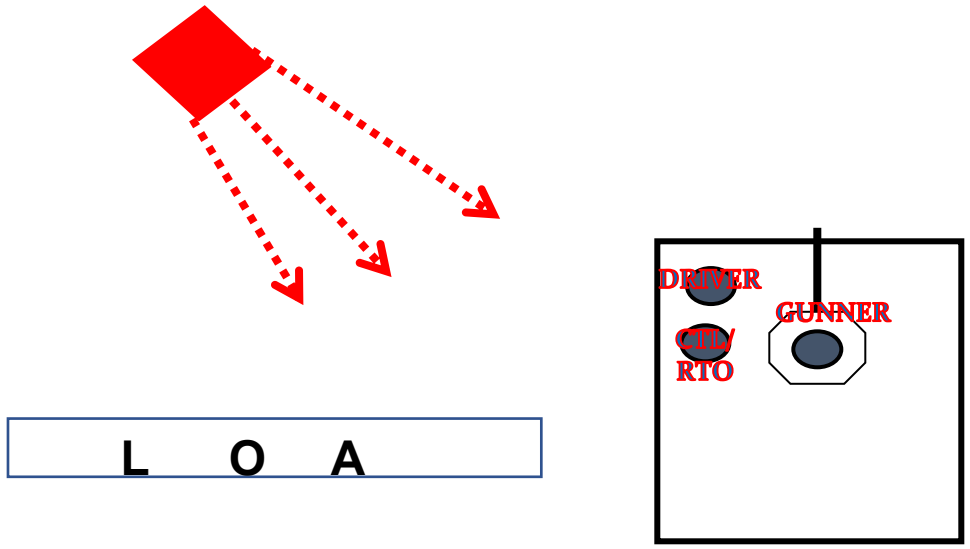
DISMOUNT RIGHT (under fire from left)



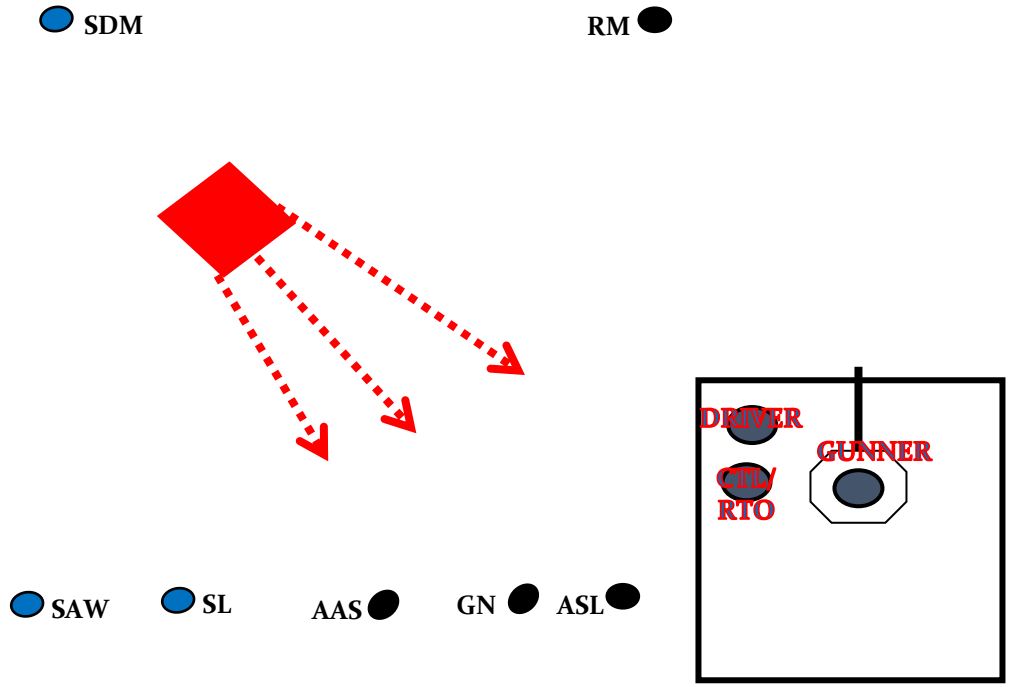
- React to contact on the left
- Extricate out of the kill zone
- Perform IAD after dismount procedures

DISMOUNT RIGHT
(under fire from left)

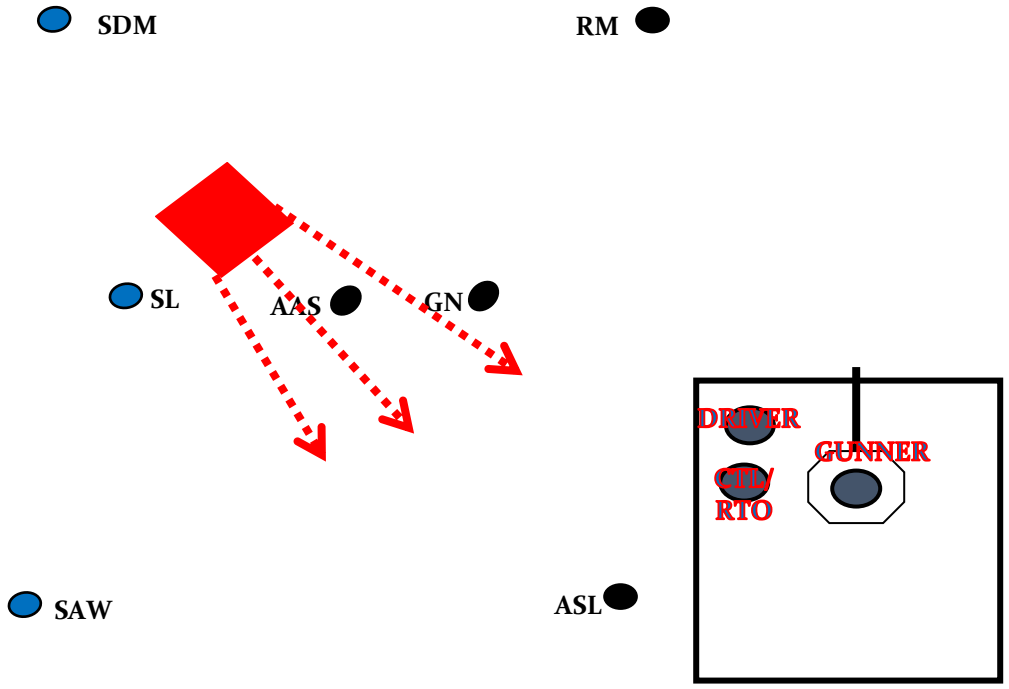
SDM SAW SL AAS GN ASL RM



DISMOUNT RIGHT
(under fire from left)



DISMOUNT RIGHT
(under fire from left)

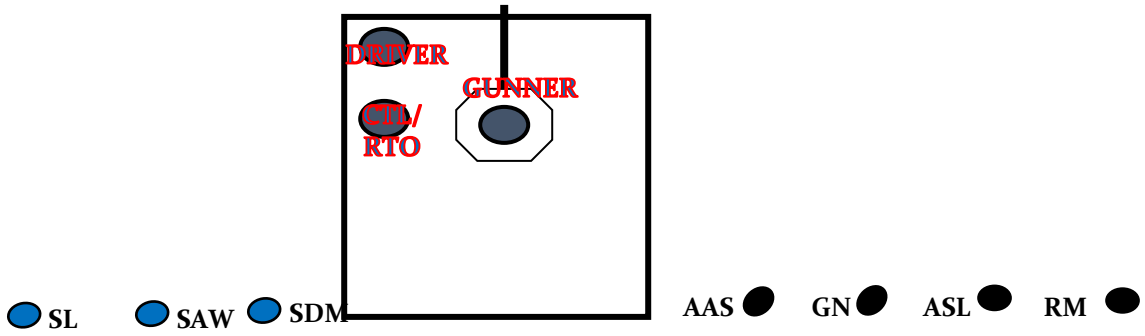


SOPs ON MOUNTING

- Provide cover while others are mounting
- Don't pile up at the door
- Move inside the AV one at a time
- Speed is of the essence
- Once inside the AV, clear weapons and muzzle down

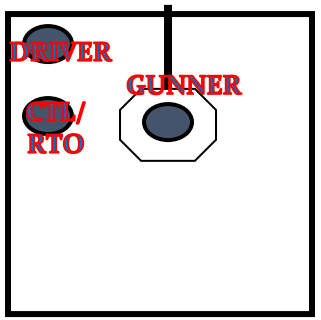


MOUNT



4
8
4
8
4
8

MOUNT FROM THE RIGHT

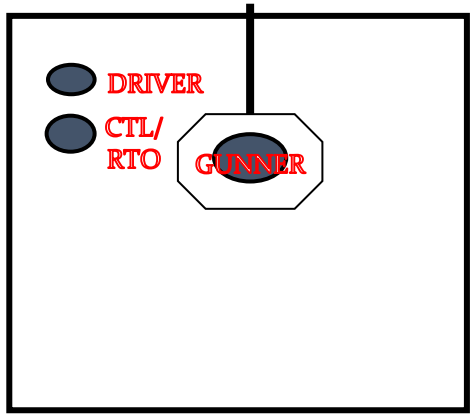


R AR SL AAS GN ASL R

4
9

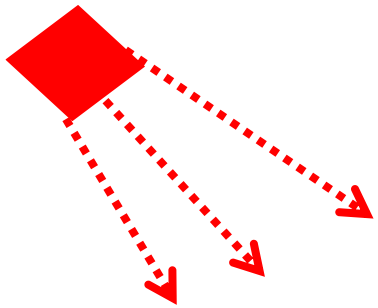
4
9
4
9

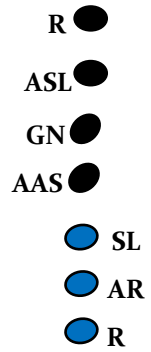
MOUNT FROM THE LEFT

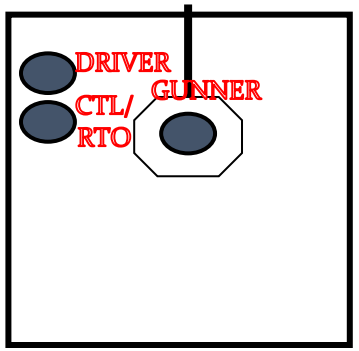


R ASL GN AAS SL AR R

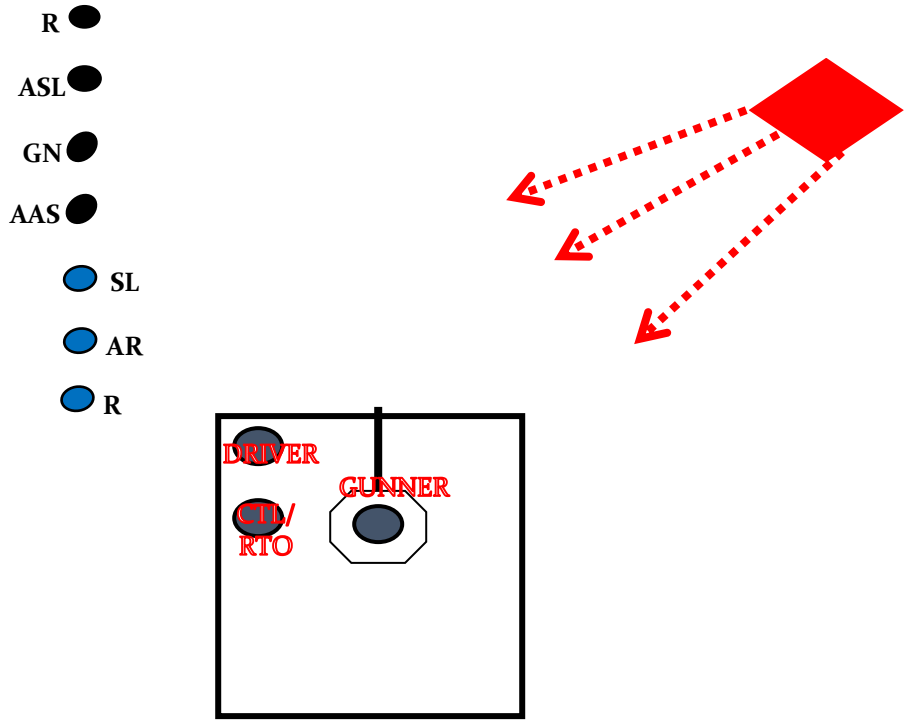
MOUNT FROM THE RIGHT
(under fire from left)







MOUNT FROM THE LEFT (under fire from right)





CAVALRY OPERATIONS

MISSION

The mission of the Cavalry unit is to perform reconnaissance and surveillance and provide security in close operations.

FUNDAMENTALS

- Facilitates the division commander's ability to maneuver divisions, brigades, and battalions and to concentrate superior combat power and apply it against the enemy at the decisive time and point

- As the commander's "eyes and ears", provides with situational awareness and enhances his ability to maneuver successfully

FUNDAMENTALS

Concentrate combat power at the decisive place and time

- Cavalry, by its role, is an **economy of force** with flexible capabilities that allow the commander to conserve the combat power of divisions or brigades for engagement where he desires

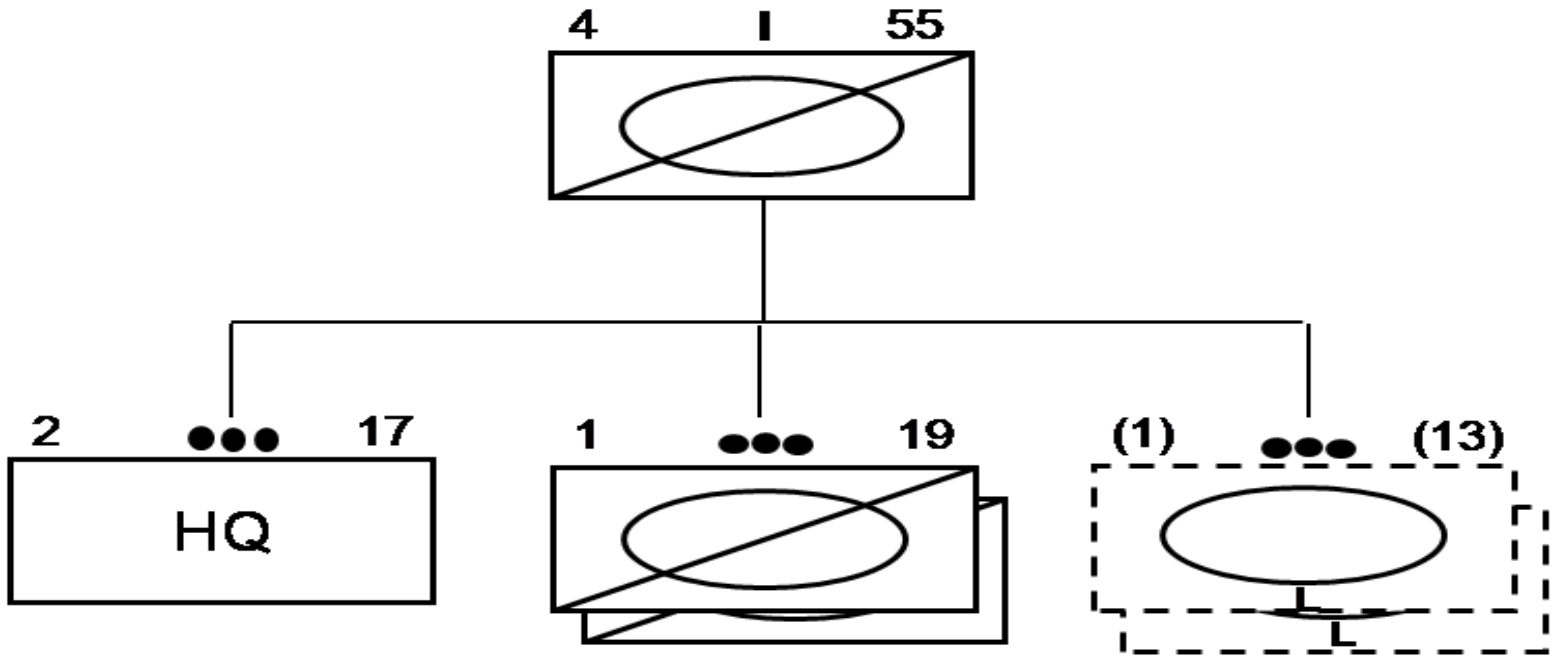
ORGANIZATION

Cavalry is organized into four (4) basic designs:

- ❑ Cavalry Squadron/Battalion
- ❑ Cavalry Troop/Company
- ❑ Air Cavalry Squadron (AAR)
- ❑ Cavalry Troop/Company (Separate)

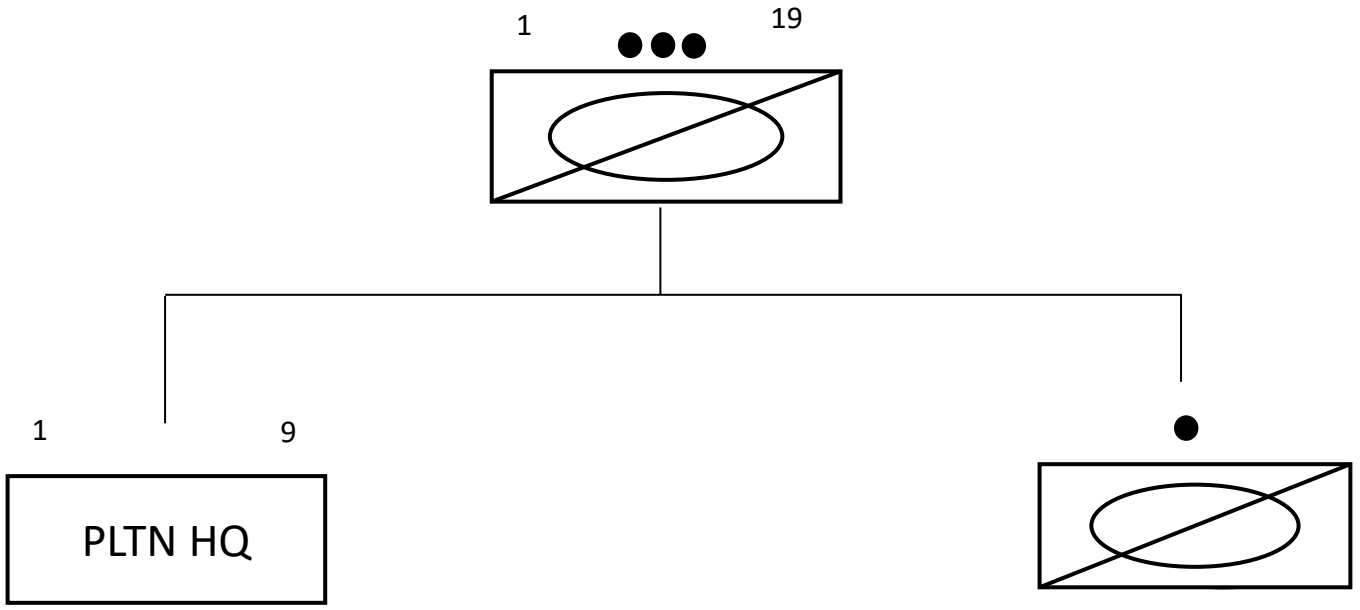
ORGANIZATION

Cavalry Troop/Company



ORGANIZATION

Scout Platoon



ORGANIZATION

Scout Section



SEC LDR SL



GNR DVR SCT LOADER



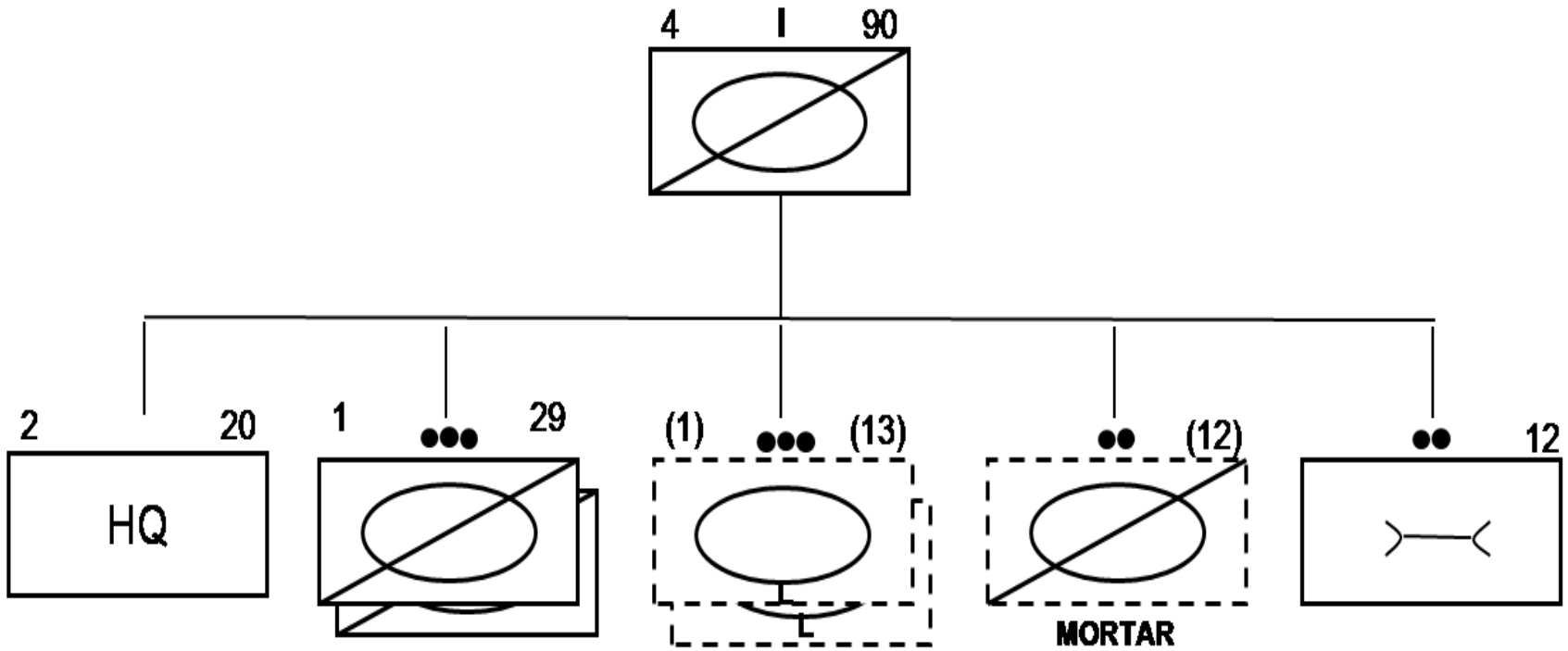
GNR DVR SCT LOADER

Cavalry Company (Separate)

- Separate cavalry troop is organized as part of separate light and armored brigades.
- It is heavy or light depending on the organization of the brigade
- It operates for the brigade commander in the operational environments for which the brigade is focused

ORGANIZATION

Cavalry Troop/Company (Separate)



CAPABILITIES

In terrain suitable for employed of cavalry units, commander can apply all their inherent combined arms mobility, firepower and shock effect to:

- Conduct reconnaissance and surveillance.
- Operate in an economy of force role without being reinforced in offensive, defensive, or delay operations
- Conduct security operations
- Move to contact independently, or as part of the advance covering force, or as the flank or rear guard of a larger force, conducted as zone reconnaissance
- Attack to overrun, penetrate, or envelop prepared enemy defenses or strong points.

LIMITATIONS

- High Logistics Requirements
- Mobility can be restricted by dense forest, steep, rugged terrain and major water obstacle, natural or man-made obstacle.



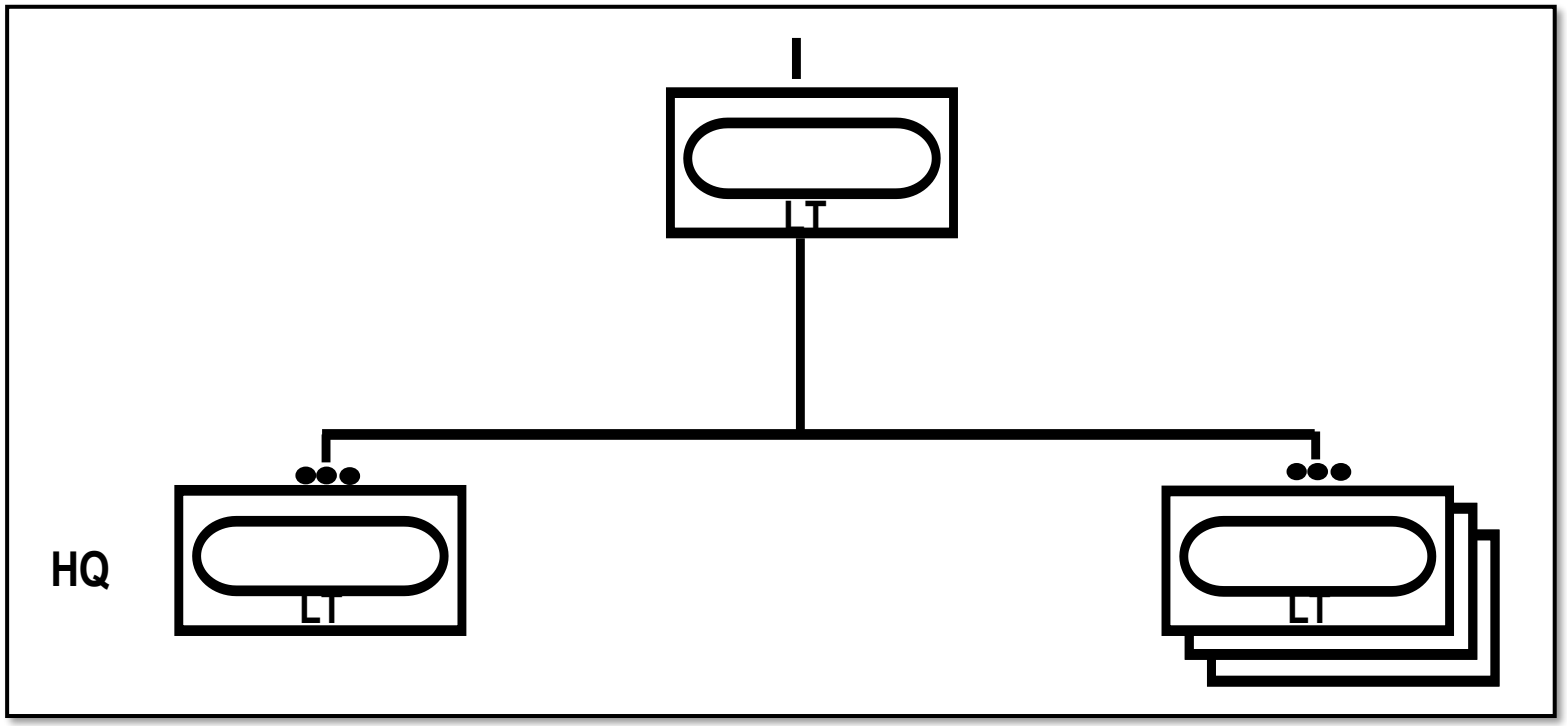
TANK OPERATIONS

MISSION

The mission of the Tank unit is to close with and neutralize enemy forces by fire, maneuver and shock effect in coordination with other arms

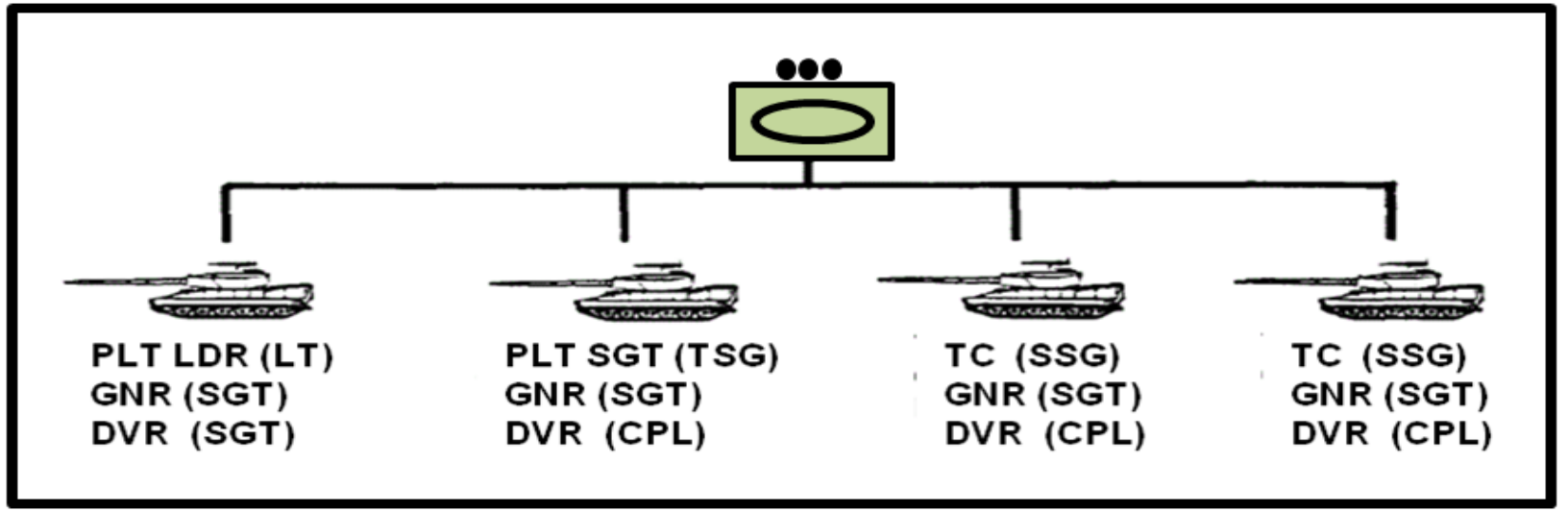
ORGANIZATION

Tank Company



ORGANIZATION

Tank Platoon



CAPABILITIES

- Support the close fight as part of a combined arms team using accurate anti-tank fires and direct fire.
- Operate on a conventional or NBC battlefield
- Use thermal sights to greatly enhance the night fighting capabilities of the combined arms team
- Operate in an opposed entry role
- Use strategic and tactical mobility to advantage
- Provide armor protection against small arms, machine gun, and overhead artillery fire.

LIMITATIONS

- The light tanks does not possess the level of protection of a main battle tank
- Tanks are vulnerable to enemy infantry with anti-armor weapons and may not withstand the impact of enemy tank fires, missiles, or antitank mines
- Tank is dependent on Flatbeds for deployment to the combat theater of operations.

LIMITATIONS

- Currently, tank requires support from the field forward support unit (FFSU) and division combat service support (CSS) elements to sustain operations.
- Infantry units do not have the organic transportation assets to support the tanks CSS needs, especially in Classes III, V, and IX.
- Mobility and firepower are restricted in extremely close terrain.

PLANNING CONSIDERATION

Mechanized Infantry

1. Where to dismount troops (On, Before, After)
2. Weapon Range
3. Objective to Combat Trains distance
4. Ex-O or F/Sgt as resource manager
5. Avenue of Approach Restrictions
6. Cannot be used against Tank Units

PLANNING CONSIDERATION

Cavalry

1. Indirect Fire Weapon Support
2. Weapon Range
3. Security Operations
4. Distance from Main Body
5. Cannot be used against Tank Units

PLANNING CONSIDERATION

Tank

1. Surface Danger Zone
2. Direct Fire Control Measures
3. SBF Positions
4. Cover Against Enemy Air
5. Can be used against other Tank or any Armored Vehicle of Enemy







MAJOR EQUIPMENT OF THE ARMOR DIVISION









M113A2 ARMORED PERSONNEL CARRIER



M113A2 APC

GENERAL DATA

Engine type	–	Detroit Diesel 6V53
Horsepower	–	212 Horsepower at 2,800 rpm
Battle Combat Weight	–	11,070 kg (24,405.1 pounds)
Net Weight	–	9,630 kg (21,230.5 pounds)
Length Overall	–	4.90 m (192.91 in)
Width	–	2.70 m (106.29 in)
Height	–	2.50 m (98.42 in)
Ground clearance (Hull)	–	0.41 m (16.14 in)
Fuel tank capacity	–	360 liters ADF
Electrical system	–	24 Volt Direct Current (VDC)
Communication	–	RF 5800 HH Harris Vehicular Radio

CAPABILITIES

Speed forward	–	66 kph (41 mph)
Speed reverse	–	14.80 kph (9.2 mph)
Speed in water	–	3 miles per hour
Cruising range (road)	–	482.80 kms (300 miles)
Fuel consumption	–	1.4 kms per liter

CHARACTERISTICS

Air transportable
Full tracked laying
Aluminum alloy
Troop or cargo carrier
Armor protected vehicle
Fully amphibious

ARMAMENTS

Machinegun – Browning M2 Heavy Barrel (HB) 12.7 mm (.50 cal) electrically or mechanically fired, left hand feed, right hand cocking by means of MK 10 charger, hand operated elevation gearbox

Maximum Effective Range	–	1,830 meters
Maximum Range	–	6,800 meters
Rate of fire	–	450 – 500 rounds/min

M113A2 REMOTE CONTROLLED WEAPON SYSTEM



M113A2 APC
Remote Controlled Weapon System (RCWS)

GENERAL DATA

Engine type	–	Detroit Diesel 6V53
Horsepower	–	212 Horsepower at 2,800 rpm
Battle Combat Weight	–	11,070 kg (24,405.1 pounds)
Net Weight	–	9,630 kg (21,230.5 pounds)
Length Overall	–	4.90 m (192.91 in)
Width	–	2.70 m (106.29 in)
Height	–	2.50 m (98.42 in)
Ground clearance (Hull)	–	0.41 m (16.14 in)
Fuel tank capacity	–	360 liters ADF
Electrical system	–	24 Volt Direct Current (VDC)
Communication	–	RF 5800 HH Harris Vehicular Radio

CAPABILITIES

Speed forward	–	66 kph (41 mph)
Speed reverse	–	14.80 kph (9.2 mph)
Speed in water	–	3 miles per hour
Cruising range (road)	–	482.80 kms (300 miles)
Fuel consumption	–	1.4 kms per liter

CHARACTERISTICS

Air transportable
Full tracked laying
Aluminum alloy
Troop or cargo carrier
Armor protected vehicle
Fully amphibious

ARMAMENT

Machinegun – Browning M2A1 .50 Cal Quick Change Barrel (QCB)
Observation using day and night cameras
Enables range measuring
Automatic Target Tracking (ATT)
Fire Control System (FCS)
Electrical Drive System (EDS)

Maximum Range	–	2,000 meters
Maximum Effective Range	–	1,700 meters
Rate of fire	–	450 – 500 rounds/min

M113A2 IFV UNMANNED TURRET SYSTEM



M113A2 IFV Unmanned Turret System (UTS)

GENERAL DATA

Engine type	–	Detroit Diesel 6V53
Horsepower	–	212 Horsepower at 2,800 rpm
Battle Combat Weight	–	11,070 kg (24,405.1 pounds)
Net Weight	–	9,630 kg (21,230.5 pounds)
Length Overall	–	4.90 m (192.91 in)
Width	–	2.70 m (106.29 in)
Height	–	2.50 m (98.42 in)
Ground clearance (Hull)	–	0.41 m (16.14 in)
Fuel tank capacity	–	360 liters ADF
Electrical system	–	24 Volt Direct Current (VDC)
Communication	–	RF 5800 HH Harris Vehicular Radio

CAPABILITIES

Speed forward	–	66 kph (41 mph)
Speed reverse	–	14.80 kph (9.2 mph)
Speed in water	–	3 miles per hour
Cruising range (road)	–	482.80 kms (300 miles)
Fuel consumption	–	1.4 kms per liter

ARMAMENTS

Primary

Cannon – 25mm Orbital Alliant Techsystem (OATK) Bush-Master MK-44 – Unmanned Turret – Day and Thermal Electro-Optic observation modules, for the gunner, a Fire Control System (FCS), which incorporates an Automatic Target Tracker (ATT), and an Electrical Drive and Stabilization System (EDSS) in elevation and traverse or mechanically fired, left hand feed, right hand cocking

Muzzle velocity: High Explosives	–	1,100 meters/sec
Armor Piercing	–	1,325 meters/sec
Authorized number of rounds	–	75 AP, 105 HE
Authorized number of rounds	–	1,840 rounds
Maximum Effective Range	–	3,000 meters
Maximum Range	–	6,800 meters
Rate of fire	–	540 – 600 rounds/min

Secondary

Machinegun – M60 7.62mm

Maximum Effective Range	–	1,100 meters
Maximum Range	–	3,725 meters
Rate of fire	–	800 rounds/min
Number of rounds in Magazine	–	460 rounds

ARMORED INFANTRY FIGHTING VEHICLE



GENERAL DATA

Engine type - - - - - --Detroit Diesel 6V53T (Turbocharged)
Horsepower- - - - - - - - - - -260 Horsepower at 2,800 rpm
Weight (Combat Load) - - - 13,687 kg (30,175.67 pounds)
Net Weight - - - - - - - - - - 11,373 kg (25,073.17 pounds)
Length Overall- - - - - - - - - - -4.90 m (192.91 in)
Width - - - - -- - - - - - - - - - 2.70 m (106.29 in)
Height - - - - -- - - - - - - - - -2.50 m (98.42 in)
Ground clearance (Hull)- - - - 0.41 m (16.14 in)
Fuel tank capacity - - - - - - - 360 liters ADF
Electrical system - - - - - - - -24 Volt Direct Current (VDC)
Communication - - - -RF 5800 HH Harris Vehicular Radio
Fuel consumption - - - - - - - - - 1.4 kms per liter

ARMAMENTS

Primary:

- Cannon – 25 mm Oerlikon KBA that can bring down low flying aircraft at a range range of 2kms and Directly attacking aircraft up to 1.5 kms.
- Rate of Fire - - - -- - - - - - - -- - - - 540-600 rounds/min
- Muzzle velocity: High Explosives - - - 1,100 meters/sec
  Armor Piercing - - - -1,325 meters/sec
- Authorized number of rounds- - - - - -75 AP, 105 HE
- Vehicle rack capacity - - - - - - - - -- - - -- - 145 rnds
- Authorized number of rounds- - - - - - - - - 1,840 rnds
- Number of rounds in EWS - - - - - - - - - - - 230 rnds
- Number of rounds stowed in AIFV - - - - - 1,610 rnds
- Max Effective Range of Target: - - - - - - - 3,000 mtrs
- Maximum Range - - - - - - - - - - - - - - - - - 6,800 mtrs

MOUNTED MORTAR SYSTEM



Mounted Mortar System (MMS)

GENERAL DATA

Engine type	–	Detroit Diesel 6V53
Horsepower	–	212 Horsepower at 2,800 rpm
Battle Combat Weight	–	11,070 kg (24,405.1 pounds)
Net Weight	–	9,630 kg (21,230.5 pounds)
Length Overall	–	4.90 m (192.91 in)
Width	–	2.70 m (106.29 in)
Height	–	2.50 m (98.42 in)
Ground clearance (Hull)	–	0.41 m (16.14 in)
Fuel tank capacity	–	360 liters ADF
Electrical system	–	24 Volt Direct Current (VDC)
Communication	–	RF 5800 HH Harris Vehicular Radio

CAPABILITIES

Speed forward	–	66 kph (41 mph)
Speed reverse	–	14.80 kph (9.2 mph)
Speed in water	–	3 miles per hour
Cruising range (road)	–	482.80 kms (300 miles)
Fuel consumption	–	1.4 kms per liter

CHARACTERISTICS

Air transportable
Full tracked Laying
Aluminum alloy
Armor protected vehicle

ARMAMENTS

Primary

81mm/120mm Mortar System
Smooth bore, muzzle loaded, self-propelled, recoiling mortar
Designed to implement improvements concluded with CARDOM combat experience
Designed to allow installation on small and medium sized vehicle
Equipped with auto-laying system incorporating an Electronic Drive System for elevation and traverse
Can be laid manually using back-up handles and an optical sight unit

Types of Ammo

81mm/120mm Mortar Bomb HE

M113A2 APC (AMBULANCE)



M113A2 APC (AMBULANCE)		
GENERAL DATA		
Engine type	–	Detroit Diesel 6V53
Horsepower	–	212 Horsepower at 2,800 rpm
Battle Combat Weight	–	11,070 kg (24,405.1 pounds)
Net Weight	–	9,630 kg (21,230.5 pounds)
Length Overall	–	4.90 m (192.91 in)
Width	–	2.70 m (106.29 in)
Height	–	2.50 m (98.42 in)
Ground clearance (Hull)	–	0.41 m (16.14 in)
Fuel tank capacity	–	360 liters ADF
Electrical system	–	24 Volt Direct Current (VDC)
Communication	–	RF 5800 HH Harris Vehicular Radio
CAPABILITIES		
Speed forward	–	66 kph (41 mph)
Speed reverse	–	14.80 kph (9.2 mph)
Speed in water	–	3 miles per hour
Cruising range (road)	–	482.80 kms (300 miles)
Fuel consumption	–	1.4 kms per liter
CHARACTERISTICS		
Air transportable		
Full tracked laying		
Aluminum alloy		
Troop or cargo carrier		
Armor protected vehicle		
Fully amphibious		

COMMANDO V-150-S



COMMANDO V-150S

GENERAL DATA

Engine type (V-150s)	–	Cummins 4 Cycle Diesel V8 (Naturally Aspirated)
(LAV-150 ST)	–	6 Cylinder in-line Diesel (Stretched Turbocharged)
Horsepower	–	202 Horsepower at 3300 rpm
Battle Combat Weight	–	10,886.21 kg (24,000 lbs)
Length Overall	–	6.14 m (242 in)
Width	–	2.36 m (89 in)
Height	–	1.98 m (78 in)
Ground clearance	–	0.64 m (25.5 in)
Fuel tank capacity	–	300 liters ADF (150 liters per tank)
Communication	–	RF 5800 HH Harris Vehicular Radio
Cruising range (road)	–	644 kms

CHARACTERISTICS

Wheeled AFV
Air transportable
Front wheel drive
Designed for internal security operations
Fully enclosed one-man turret
Fully amphibious
Run flat tire

ARMAMENTS

Primary

Machinegun – Browning M2 Heavy Barrel (HB) 12.7 mm (.50 cal) electrically or mechanically fired, left hand feed, right hand cocking by means of MK 10 charger, hand operated elevation gearbox

Maximum Effective Range	–	1,830 meters
Maximum Range	–	6,800 meters
Rate of fire	–	450 – 500 rounds/min

Secondary

Machinegun	–	M60 7.62 mm
Maximum Effective Range	–	1,100 meters
Maximum Range	–	3,725 meters
Rate of fire	–	900 rounds/min

SIMBA FIGHTING VEHICLE



SIMBA FIGHTING VEHICLE

GENERAL DATA

Engine type	–	Perkins phaser 6 ltr
Engine series	–	210 TI Turbo-charged intercooled Diesel 6 Cylinder in-line
Horsepower	–	212 Horsepower at 2500 rpm
Battle Combat	–	10160.46 kg (11.2 tons)
Length Overall	–	5.35 m (210.62 in)
Width	–	2.50 m (96.42 in)
Height	–	2.19 m (86.22 in)
Ground clearance (Hull)	–	0.45 m (18 in)
Ground clearance (Axle)	–	0.27 m (11 in)
Fuel tank capacity	–	296 liters ADF
Communication	–	RF 5800 HH Harris Vehicular Radio

CAPABILITIES

Maximum Speed- - - - - - - - - - 100 kph (62.5 mph)
Maximum vertical obstacle- - - 0.43 m (17 in)
Cruising range (road) - - - - - - 660 kms (409.2 miles)

CHARACTERISTICS

Armored Monocoque hull
Wheeled AFV
Two or Four wheel drive
Air conditioned

ARMAMENTS

Machinegun – Browning M2 Quick Change Barrel (QCB) 12.7 mm (.50 cal) electrically or mechanically fired, left hand feed, right hand cocking by means of MK 10 charger, hand operated elevation gearbox

Maximum Effective Range - - - - - - - - - - 1,830 meters
Maximum Range - - - - - - -- - - - - - - - - - 6,800 meters
Rate of fire - 450 – 500 rounds/min

FIRE SUPPORT VEHICLE (FSV)



Fire Support Vehicle (FSV)

GENERAL DATA

Engine type	–	Detroit Diesel 6V53
Horsepower	–	212 Horsepower at 2,800 rpm
Battle Combat Weight	–	11,070 kg (24,405.1 pounds)
Net Weight	–	9,630 kg (21,230.5 pounds)
Length Overall	–	4.90 m (192.91 in)
Width	–	2.70 m (106.29 in)
Height	–	2.50 m (98.42 in)
Ground clearance (Hull)	–	0.41 m (16.14 in)
Fuel tank capacity	–	360 liters ADF
Electrical system	–	24 Volt Direct Current (VDC)
Communication	–	RF 5800 HH Harris Vehicular Radio

CAPABILITIES

Speed forward	–	66 kph (41 mph)
Speed reverse	–	14.80 kph (9.2 mph)
Speed in water	–	3 miles per hour
Cruising range (road)	–	482.80 kms (300 miles)
Fuel consumption	–	1.4 kms per liter

ARMAMENTS

Primary

Main Gun – Ordnance 76mm L23A1

Maximum Effective Range (Direct Fire)	–	2,200 meters
Maximum Range (Semi-Direct Fire)	–	5,000 meters
Maximum Ammo Load	–	40 rounds

Six smoke grenade discharger

Types of Ammo

Direct Lay			Semi - Direct Lay		
HESH	–	2,200 meters	HESH	–	5,000 meters
PRAC/SH	–	2,200 meters	PRAC/SH	–	5,000 meters
PRACTICE	–	2,200 meters	HE	–	5,000 meters
CANISTER	–	150 meters	PRACTICE	–	5,000 meters

Secondary

Machinegun – M60 7.62mm

Maximum Effective Range	–	1,100 meters
Maximum Range	–	3,725 meters
Rate of fire	–	800 rounds/min
Number of rounds in Magazine	–	460 rounds

SABRAH LIGHT TANK ASCOD II



Performance Specification:

COMBAT WEIGHT	- 30 Tons
LENGTH	- 8.368 Meters
WIDTH	- 2.786 Meters
HEIGHT	- 3.67 Meters
CREW	- 3
ARMOR PROTECTION AND CREW SAFETY	- Highly Sophisticated modular armor protection concept with armored steel hull - Level 4 ballistic protection per STANAG 4569
MAIN ARMAMENT	- 105mm 52 calibers, coiled, low recoil - Combat effective range, 3,600 meters 6 rounds per minute
CO-AXIAL	- 7.62mm machine gun
SMOKE/GRENADE LAUNCHERS	- Eight (8) 76mm Smoke/Grenade Launchers
ENGINE	- Diesel with 8-v cylinders, 4-stroke direct injection, Turbo-charger and air cooler, compatible with NATO fuels
MAXIMUM SPEED	- 72 km/hr (road)
COMMUNICATION	- Radio - BMS and Navigation System

GUARANI 6X6 WHEELED APC



6x6 WHEELED APC Specifications	
Date Acquired	2024 under Horizon 2 Project
Max Speed	72 Kph
Engine	Cummins ISC diesel engine
Primary	Main Gun 12.7mm HMG
Secondary	Co-axial Gun 7.62mm GPMG
Maximum effective range	1,830 meters
Maximum range	6,800 meters
Rate of Fire	485 – 635 rds/min
Crew	3 + 7 Passenger
Number of Units	5

Summary

- ❑Fundamentals of Armor Operations
- ❑Characteristics and Organization of Armor Units
- ❑Armor Operation Planning Considerations
- ❑Armor Division Major Equipment

Outcome

By this time the students will be able to discuss the Fundamentals of Armor Operations in accordance with the Armor Operations Manual.

QUESTIONS






HAVE A
GREAT DAY

End of Presentation